BTS (C) – III – 07 – 067 (B)

B. Tech Degree III Semester Examination, November 2007

IT/CS 303 DISCRETE COMPUTATIONAL STRUCTURES

(2006 Admissions)

Time : 3 Hours   Maximum Marks : 100

PART A

(Answer ***ALL*** questions)
(All questions carry ***FIVE*** marks)

(8 x 5 = 40)

I (a) Find the truth value of the proposition $p \wedge (q \rightarrow r)$.

(b). Use Principle of Mathematical Induction to show that

$$1^2 + 2^2 + 3^2 + \ldots\ldots\ldots\ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6} \text{ for all } n \in N.$$

(c) Write the algorithm for finding the maximum value in a sequence.

(d) Solve the recurrence relation $S_n = 2S_{n-1}$ subject to the initial condition $S_o = 1$.

(e) Define a connected graph. Draw the complete graph K_4 and examine whether it is a connected graph.

(f) Show that the sum of degrees of all the vertices in a graph G is even..

(g) Consider the algebraic system $(\{0,1\}, *)$ where $*$ is a multiplication operation. Determine whether $(\{0,1\}, *)$ is a semi group.

(h) Let L be a lattice. Then for every a and b show that

   (i) $a \vee b = b$ if and only if $a \leq b$

   (ii) $a \wedge b = a$ if and only if $a \leq b$. → $b \geq a \wedge b$ ($\because a \wedge b = a$)

$\therefore b \geq a$

Then converse easily

GLB & LUB

PART – B

(All questions carry ***FIFTEEN*** marks)

(4 x 15 = 60)

I. (a) Using truth table show that $\neg(p \rightarrow q) \rightarrow p$.

(b) Show that the negation of $p \rightarrow q$ is logically equivalent to $p \wedge \neg q$.

OR

II. (a) Define an equivalence relation. If R is an equivalence relation, show that R^{-1} is also an equivalence relation.

(b) Let $A = B = R$, the set of real numbers. Let $f : A \rightarrow B$ be given by the formula $f(x) = 2x^3 - 1$. Show that f is a bijective function. Find its inverse.

III. (a) Show that $\lg n! = \Theta(n \lg n)$.

(b) Solve the recurrence relation $a_n = a_{n-1} + 3$ subject to the initial condition $a_1 = 2$.

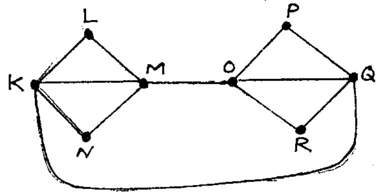
OR

IV. How many permutations can be made out of the letters of the word 'COMPUTER'? How many of these

   (i) begin with C?
   (ii) end with R?
   (iii) begin with C and end with R?
   (iv) C and R occupy the end places?

V. (a) Show that in any graph G, there are an even number of vertices of odd degree.

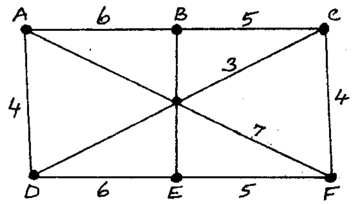
(b) Define Euler path and circuits. Use Fleury's algorithm to find an Euler Circuit in the following graph.



OR

VI. (a) Define a minimal spanning tree.

(b) Use Kruskal's Algorithm to find minimal spanning tree of the weighted graph shown in the figure.



VII. (a) Define a Lattice. Draw the Hasse diagram for the divisibility relation on the set $A = \{2, 3, 6, 12, 24, 36\}$. Check whether it is a Lattice.

(b) Let G be a set of all non zero real numbers and let $a * b = \dfrac{ab}{2}$. Show that $(G, *)$ is an abelian group.

OR

VIII. (a) Let $(S, *)$ and $(T, *')$ be monoids with identities e and e' respectively. Let $f : S \to T$ be an isomorphism. Then show that $f(e) = e'$.

(b) Determine whether the posets shown in the figure are Lattices or not.

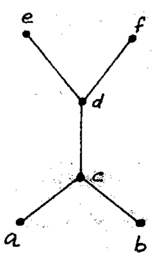


Fig. (1)

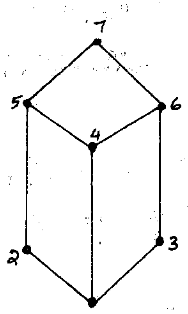


Fig. (2)

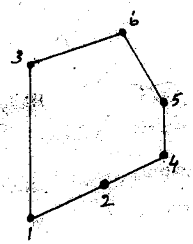


Fig. (3)

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